

# Chapter 2: Atomic Structure & Interatomic Bonding



## ISSUES TO ADDRESS...

- What promotes bonding?
- What types of bonds are there?
- What properties are inferred from bonding?

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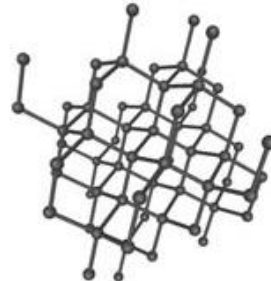
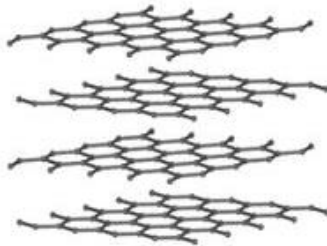
# Why we study atomic structure and interatomic bonding

- They explain a material's properties
- Consider Carbon (C)



*Dull, opaque, soft, common*

*Brilliant, transparent, hard, rare*



*Some of the important properties of solid materials depend on geometrical atomic arrangements, and also the interactions that exist among constituent atoms or molecules*

# Atomic Structure (Freshman Chem.)



- atom –  
    electrons –  $9.11 \times 10^{-31}$  kg  
    protons    }  
    neutrons   } $1.67 \times 10^{-27}$  kg
- atomic number = # of protons in nucleus of atom  
                  = # of electrons of neutral species  
                  (for an electrically neutral or  
                  complete atom)

- A [=] atomic mass unit = amu =  $1/12$  mass of  $^{12}\text{C}$

Atomic wt = wt of  $6.023 \times 10^{23}$  molecules or atoms

$$1 \text{ amu/atom} = 1\text{g/mol}$$

C 12.011  
H 1.008 etc.



The *atomic mass* ( $A$ ) of a specific atom may be expressed as the sum of the masses of protons and neutrons within the nucleus. Although the number of protons is the same for all atoms of a given element, the number of neutrons ( $N$ ) may be variable. Thus atoms of some elements have two or more different atomic masses, which are called **isotopes**. The **atomic weight** of an element corresponds to the weighted average of the atomic masses of the atom's naturally occurring isotopes.<sup>2</sup> The **atomic mass unit (amu)** may be used to compute atomic weight. A scale has been established whereby 1 amu is defined as  $\frac{1}{12}$  of the atomic mass of the most common isotope of carbon, carbon 12 ( $^{12}\text{C}$ ) ( $A = 12.00000$ ). Within this scheme, the masses of protons and neutrons are slightly greater than unity, and

$$A \cong Z + N \quad (2.1)$$



# Atomic Structure

- Valence electrons determine all of the following properties

- 1) Chemical

- 2) Electrical

- 3) Thermal

- 4) Optical

- 5) Mechanical

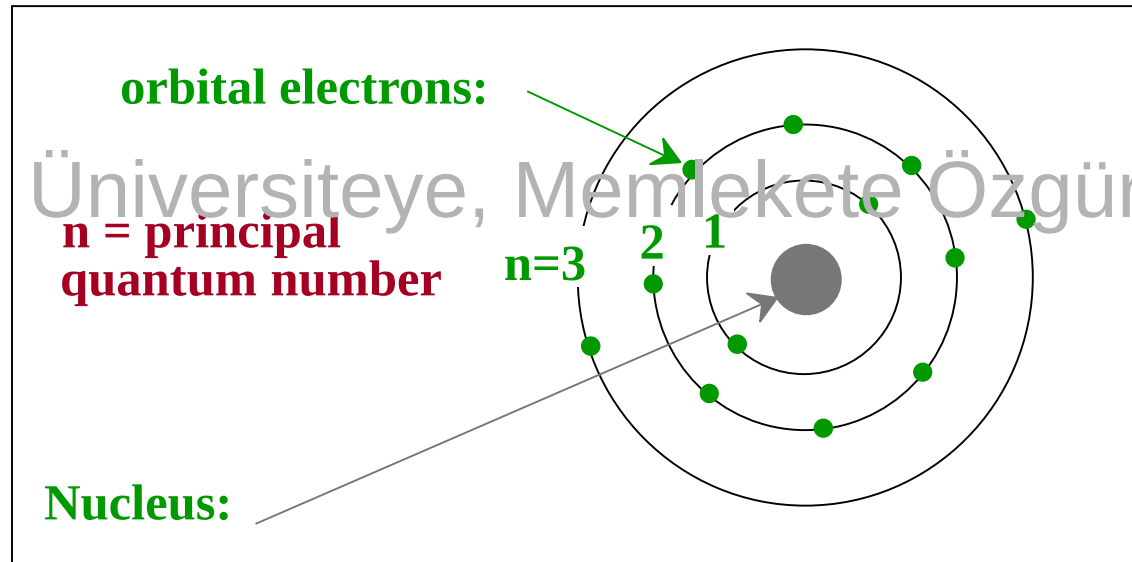
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# Importance of Electronic Structure

- **Outer (*valence*) electrons easiest to remove, causing the atom to be ionized**
- **Inner electrons tightly bound**
- **Valence electrons are the ones participating in bonding**
- **Quantum Mechanics describes allowed energy states**

# Atomic Structure

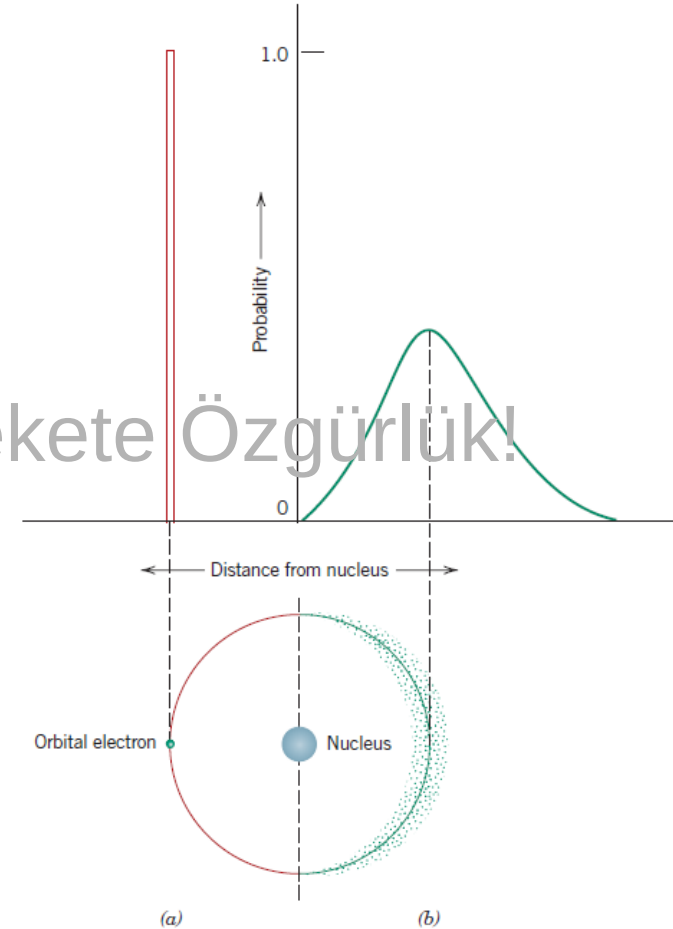
- **Bohr model:**
  - Electrons orbiting the massive nucleus
  - Nucleus contains protons (charged) & neutrons (uncharged)



- Principal quantum numbers represent solutions to Schrödinger's equation
- $Z = \# \text{ protons}$ ,  $N = \# \text{ neutrons}$ , Atomic Mass =  $A = Z + N$

# Electronic Structure

- Electrons have wavelike and particulate properties.
  - This means that electrons are in **orbitals** defined by a probability.
  - Each orbital at discrete energy level is determined by **quantum numbers (4)**. The size, shape, and spatial orientation of an electron's probability density are specified by three of these quantum numbers.



Comparison of the (a) Bohr and (b) wavemechanical atom models in terms of electron distribution.



# Electronic Structure

## Quantum #

$n$  = principal (energy level-shell)

$l$  = subsidiary (orbitals) (2nd q. n.)

$m_l$  = magnetic

$m_s$  = spin moment (one for each spin orient.)  $\frac{1}{2}, -\frac{1}{2}$

## Designation

$K, L, M, N, O$  (1, 2, 3, etc.)

$s, p, d, f$  (0, 1, 2, 3, ...,  $n-1$ )

1, 3, 5, 7 ( $-l$  to  $+l$ )

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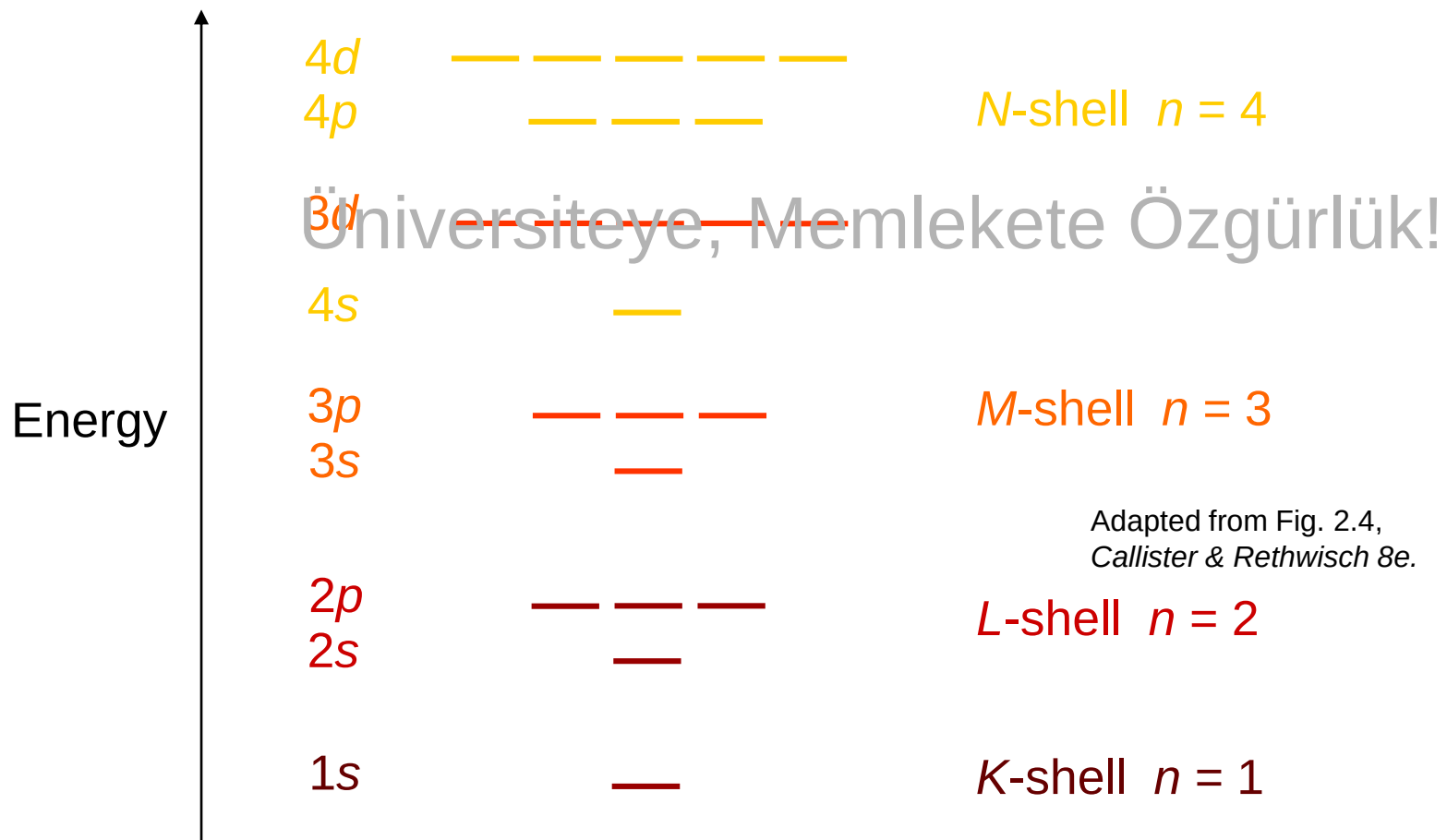
**Table 2.1** The Number of Available Electron States in Some of the Electron Shells and Subshells

| Principal Quantum Number $n$ | Shell Designation | Subshells | Number of States | Number of Electrons |           |
|------------------------------|-------------------|-----------|------------------|---------------------|-----------|
|                              |                   |           |                  | Per Subshell        | Per Shell |
| 1                            | $K$               | $s$       | 1                | 2                   | 2         |
| 2                            | $L$               | $s$       | 1                | 2                   | 8         |
|                              |                   | $p$       | 3                | 6                   |           |
| 3                            | $M$               | $s$       | 1                | 2                   | 18        |
|                              |                   | $p$       | 3                | 6                   |           |
|                              |                   | $d$       | 5                | 10                  |           |
|                              |                   | $f$       | 7                | 14                  |           |
| 4                            | $N$               | $s$       | 1                | 2                   | 32        |
|                              |                   | $p$       | 3                | 6                   |           |
|                              |                   | $d$       | 5                | 10                  |           |
|                              |                   | $f$       | 7                | 14                  |           |

# Electron Energy States

Electrons...

- have discrete **energy states**
- tend to occupy lowest available energy state.



# SURVEY OF ELEMENTS

- Most elements: Electron configuration **not stable**.

| Element   | Atomic # | Electron configuration                                |
|-----------|----------|-------------------------------------------------------|
| Hydrogen  | 1        | $1s^1$                                                |
| Helium    | 2        | $1s^2$ (stable)                                       |
| Lithium   | 3        | $1s^2 2s^1$                                           |
| Beryllium | 4        | $1s^2 2s^2$                                           |
| Boron     | 5        | $1s^2 2s^2 2p^1$                                      |
| Carbon    | 6        | $1s^2 2s^2 2p^2$                                      |
| ...       | ...      | ...                                                   |
| Neon      | 10       | $1s^2 2s^2 2p^6$ (stable)                             |
| Sodium    | 11       | $1s^2 2s^2 2p^6 3s^1$                                 |
| Magnesium | 12       | $1s^2 2s^2 2p^6 3s^2$                                 |
| Aluminum  | 13       | $1s^2 2s^2 2p^6 3s^2 3p^1$                            |
| ...       | ...      | ...                                                   |
| Argon     | 18       | $1s^2 2s^2 2p^6 3s^2 3p^6$ (stable)                   |
| ...       | ...      | ...                                                   |
| Krypton   | 36       | $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$ (stable) |

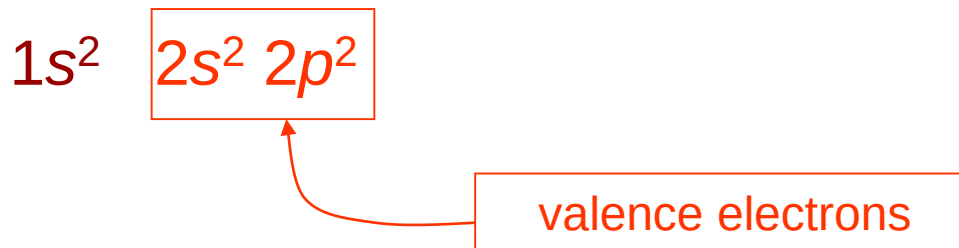
Adapted from Table 2.2,  
Callister & Rethwisch 8e.

- Why? **Valence** (outer) shell usually not filled completely.

# Electron Configurations

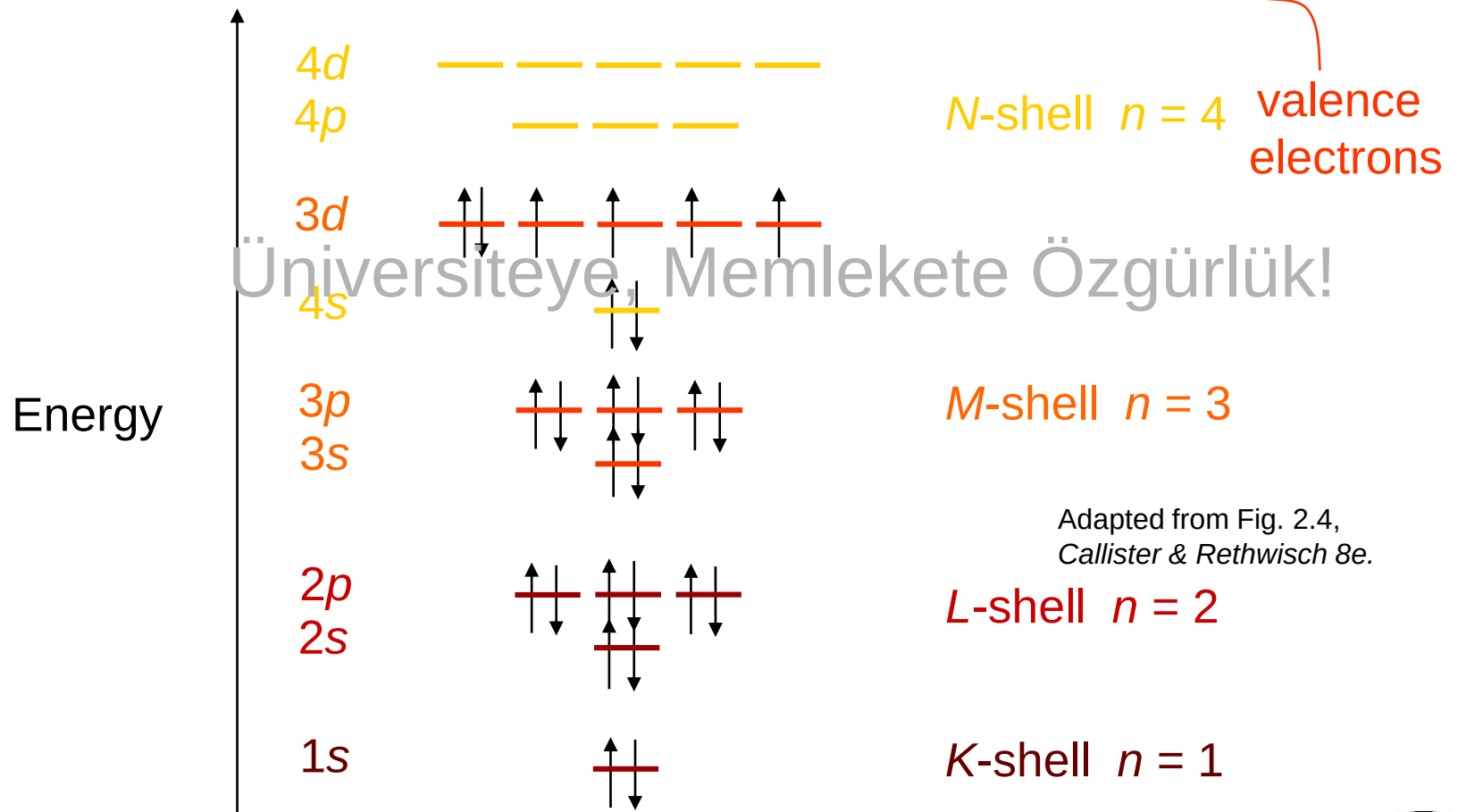
- **Valence electrons** – those in unfilled shells
  - Filled shells more stable
  - Valence electrons are most available for bonding and tend to control the chemical properties
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– example: C (atomic number = 6)



# Electronic Configurations

ex: Fe - atomic # = 26  $1s^2 2s^2 2p^6 3s^2 3p^6$   $3d^6 4s^2$



Adapted from Fig. 2.4,  
Callister & Rethwisch 8e.

# Subshells



- Each quantum level ( $n=1, 2, 3\dots$ ) has sublevels created by differences in electron spins
- Subshells called s, p, d, f

- Nomenclature:

(principal quantum number) (subshell) (# within that subshell)

*e.g.*  $2p^6$



# Stable Electron Configurations

- Inert elements....
  - Have complete s and p subshells
  - Tend to be nonreactive

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| Z  | Element | Configuration                                |
|----|---------|----------------------------------------------|
| 2  | He      | $1s^2$                                       |
| 10 | Ne      | $1s^2 2s^2 2p^6$                             |
| 18 | Ar      | $1s^2 2s^2 2p^6 3s^2 3p^6$                   |
| 36 | Kr      | $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$ |

# Limited Capacity in Shells/Subshells

- Subshells are limited to max # of electrons:

- $s$  has 2 electrons max
- $p$  has 6 electrons max
- $d$  has 10 electrons max
- $f$  has 14 electrons max

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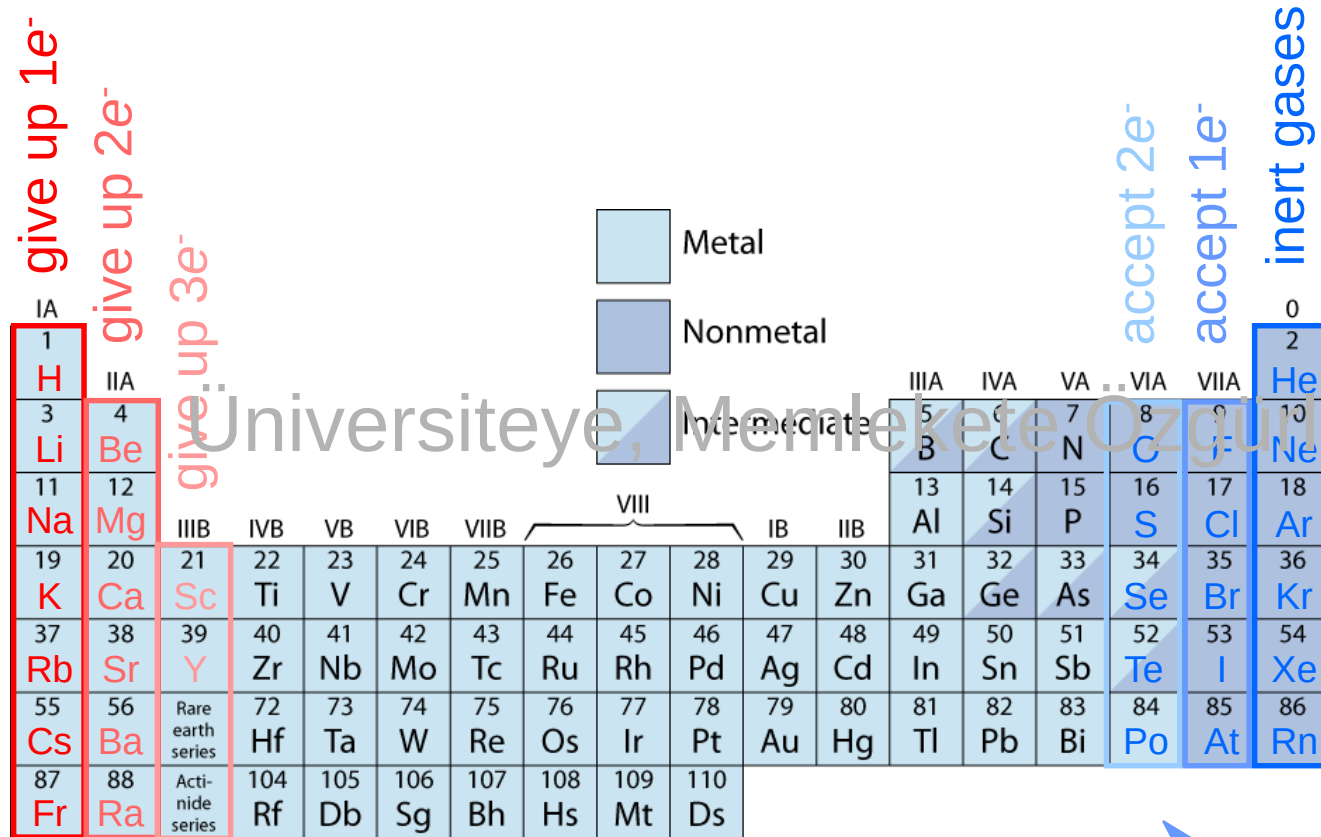
- Shells are limited to max # of subshells

- $N=1$  has only  $s$
- $N=2$  has only  $s$  &  $p$
- $N=3$  has only  $s$ ,  $p$  &  $d$
- $N=4$  has  $s$ ,  $p$ ,  $d$ , &  $f$



# The Periodic Table

- Columns: Similar **Valence** Structure



| IA       | IIA      | Intermediate      |           |           |           |           |           |           |           |          |          | IIIA     | IVA      | VA       | VIA      | VIIA     | 0        |
|----------|----------|-------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----------|----------|----------|----------|----------|----------|----------|----------|
| 1<br>H   | 2<br>He  |                   |           |           |           |           |           |           |           |          |          | 3<br>B   | 4<br>C   | 5<br>N   | 6<br>O   | 7<br>F   | 8<br>Ne  |
| 3<br>Li  | 4<br>Be  |                   |           |           |           |           |           |           |           |          |          | 13<br>Al | 14<br>Si | 15<br>P  | 16<br>S  | 17<br>Cl | 18<br>Ar |
| 11<br>Na | 12<br>Mg | 21<br>Sc          | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu | 30<br>Zn | 31<br>Ga | 32<br>Ge | 33<br>As | 34<br>Se | 35<br>Br | 36<br>Kr |
| 19<br>K  | 20<br>Ca | 39<br>Y           | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag | 48<br>Cd | 49<br>In | 50<br>Sn | 51<br>Sb | 52<br>Te | 53<br>I  | 54<br>Xe |
| 37<br>Rb | 38<br>Sr | Rare earth series | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au | 80<br>Hg | 81<br>Tl | 82<br>Pb | 83<br>Bi | 84<br>Po | 85<br>At | 86<br>Rn |
| 55<br>Cs | 56<br>Ba | Actinide series   | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds |          |          |          |          |          |          |          |          |
| 87<br>Fr | 88<br>Ra |                   |           |           |           |           |           |           |           |          |          |          |          |          |          |          |          |

Adapted from  
Fig. 2.6,  
Callister &  
Rethwisch 8e.

Electropositive elements:  
Readily give up electrons  
to become + ions.

Electronegative elements:  
Readily acquire electrons  
to become - ions.

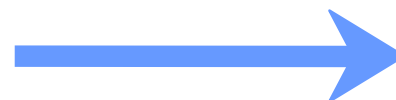
# Electronegativity

- Ranges from 0.7 to 4.0,
- Large values: tendency to acquire electrons.

|     |     |         |     |     |     |      |      |     |     |     |     |      |     |     |     |      |    |
|-----|-----|---------|-----|-----|-----|------|------|-----|-----|-----|-----|------|-----|-----|-----|------|----|
| IA  |     |         |     |     |     |      |      |     |     |     |     |      |     |     |     |      | 0  |
| H   |     |         |     |     |     |      |      |     |     |     |     |      |     |     |     |      | He |
| 2.1 | IIA |         |     |     |     |      |      |     |     |     |     | IIIA | IVA | VA  | VIA | VIIA | -  |
| Li  | Be  |         |     |     |     |      |      |     |     |     |     | B    | C   | N   | O   | F    | Ne |
| 1.0 | 1.5 |         |     |     |     |      |      |     |     |     |     | 2.0  | 2.5 | 3.0 | 3.5 | 4.0  | -  |
| Na  | Mg  |         |     |     |     |      |      |     |     |     |     | Al   | Si  | P   | S   | Cl   | Ar |
| 0.9 | 1.2 | IIIB    | IVB | VB  | VIB | VIIA | VIII |     |     | IB  | IIB | 1.5  | 1.8 | 2.1 | 2.5 | 3.0  | -  |
| K   | Ca  | Sc      | Ti  | V   | Cr  | Mn   | Fe   | Co  | Ni  | Cu  | Zn  | Ga   | Ge  | As  | Se  | Br   | Kr |
| 0.8 | 1.0 | 1.3     | 1.5 | 1.6 | 1.6 | 1.5  | 1.8  | 1.8 | 1.8 | 1.9 | 1.6 | 1.6  | 1.8 | 2.0 | 2.4 | 2.8  | -  |
| Rb  | Sr  | Y       | Zr  | Nb  | Mo  | Tc   | Ru   | Rh  | Pd  | Ag  | Cd  | In   | Sn  | Sb  | Te  | I    | Xe |
| 0.8 | 1.0 | 1.2     | 1.4 | 1.6 | 1.8 | 1.9  | 2.2  | 2.2 | 2.2 | 1.9 | 1.7 | 1.7  | 1.8 | 1.9 | 2.1 | 2.5  | -  |
| Cs  | Ba  | La-Lu   | Hf  | Ta  | W   | Re   | Os   | Ir  | Pt  | Au  | Hg  | Tl   | Pb  | Bi  | Po  | At   | Rn |
| 0.7 | 0.9 | 1.1-1.2 | 1.3 | 1.5 | 1.7 | 1.9  | 2.2  | 2.2 | 2.2 | 2.4 | 1.9 | 1.8  | 1.8 | 1.9 | 2.0 | 2.2  | -  |
| Fr  | Ra  | Ac-No   |     |     |     |      |      |     |     |     |     |      |     |     |     |      |    |
| 0.7 | 0.9 | 1.1-1.7 |     |     |     |      |      |     |     |     |     |      |     |     |     |      |    |



Smaller electronegativity



Larger electronegativity

Adapted from Fig. 2.7, Callister & Rethwisch 8e. (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.

# Atomic Bonding

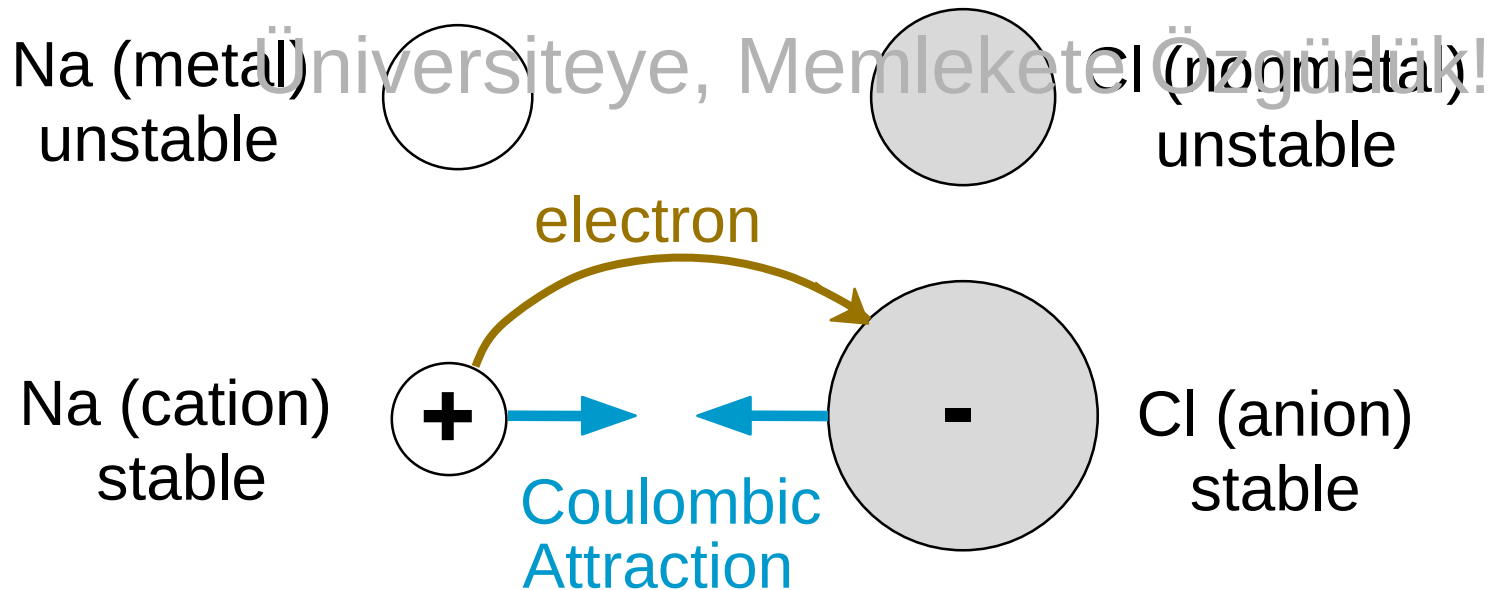
- **Defines structure and properties of materials**
  - **Types of bonding:**
    - **Ionic**
    - **Covalent**
    - **Metallic**
    - **Van der Waals**
    - **Hydrogen**
- Primary bonding (strong)*
- Secondary bonding (weak)*

# Electrons and Bonding

- **Bonding of adjacent atoms**
- **Transfer of electrons**
  - **Ionic bonds**
- **Sharing of Electrons**
  - **Metallic bonds**
  - **Covalent bonds**
- **Secondary Mechanisms**
  - **Hydrogen bonds**
  - **Van der Waals**
  - **Dipole interactions**

# Ionic Bonding

- Occurs between + and - ions.
- Requires **electron transfer**.
- Large difference in electronegativity required.
- Example: NaCl



Ionic bond – metal + nonmetal

↑  
donates  
electrons

↑  
accepts  
electrons

Dissimilar electronegativities

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ex: MgO

Mg  $1s^2 2s^2 2p^6 3s^2$   
[Ne]  $3s^2$

O  $1s^2 2s^2 2p^4$

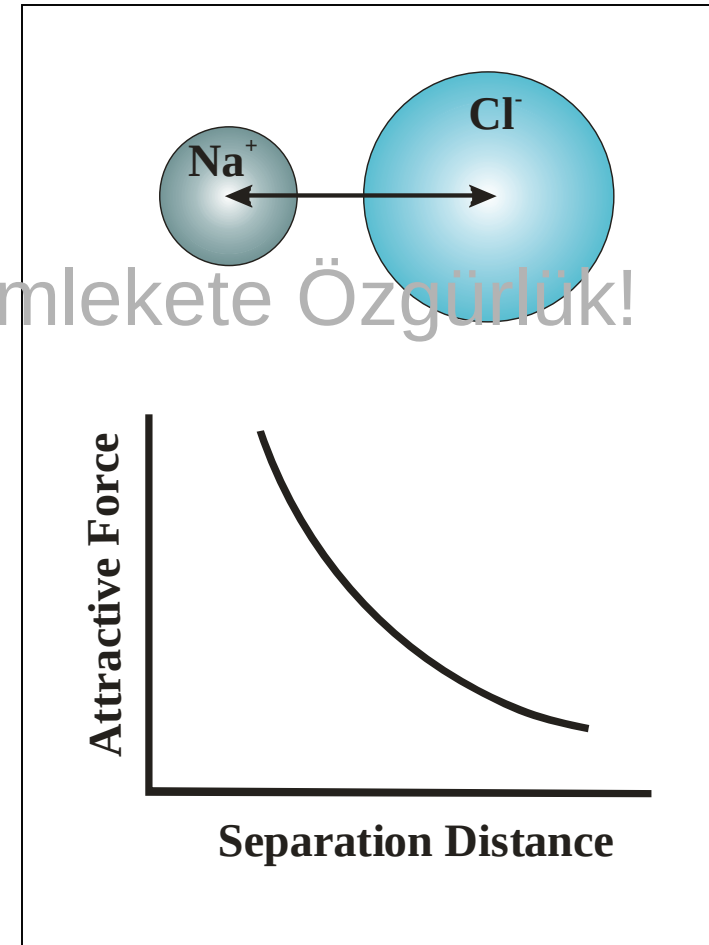
$Mg^{2+}$   $1s^2 2s^2 2p^6$   
[Ne]

$O^{2-}$   $1s^2 2s^2 2p^6$   
[Ne]

# Coulomb Force

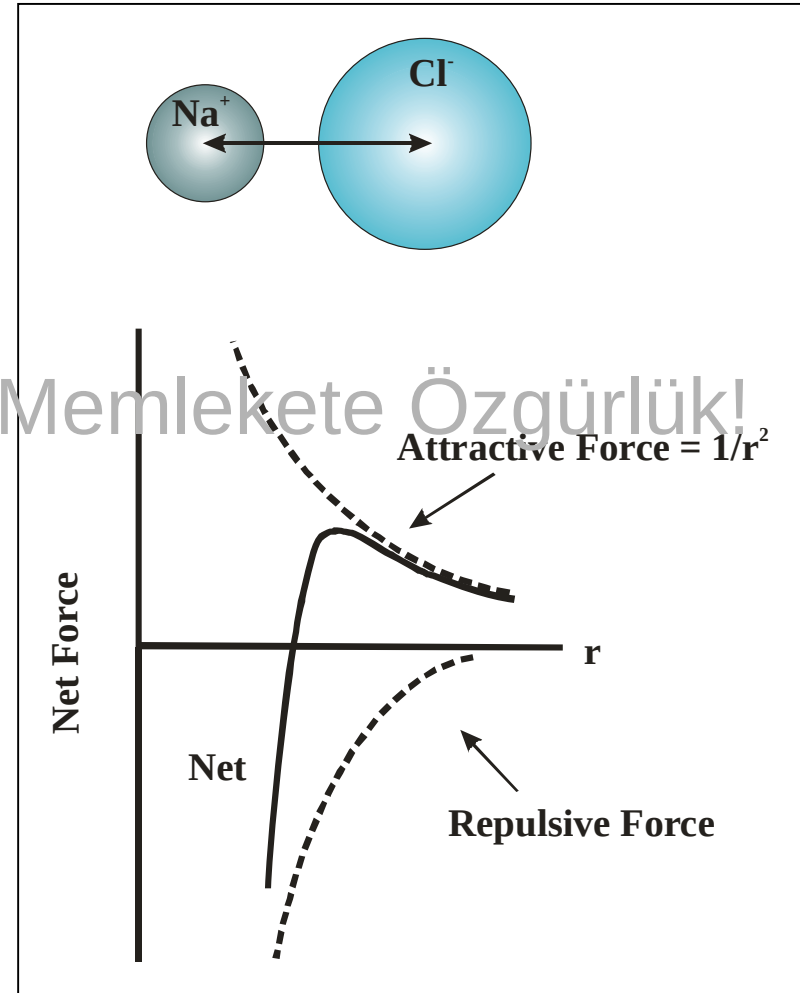
$$F(r) \propto \frac{(q_1 q_2)}{r^2}$$

- **Coulombic attraction between oppositely charged atoms**
- **Attraction proportional to  $1/r^2$**
- **Counteracting force by overlap of electron clouds and charged nuclei ( $F = \exp(-a/r)$ )**



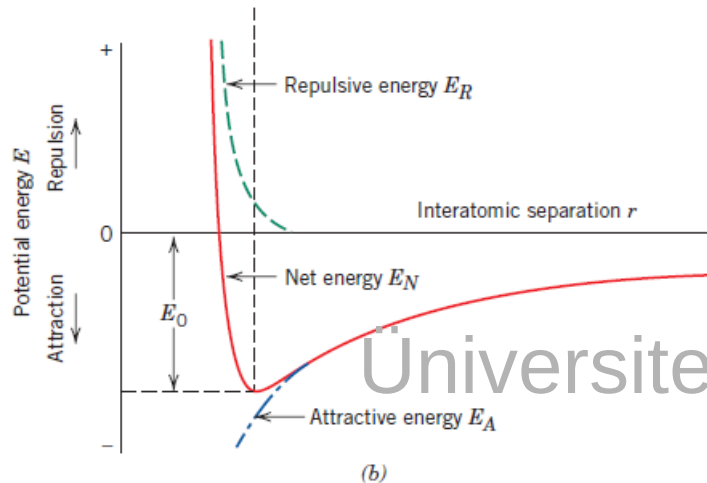
# Equilibrium Separation Distance

- **Attractive and repulsive forces balance at same distance**
- **This distance is the equilibrium separation distance**
- **Note that pulling or pushing the atoms apart causes a restoring force**





# Potential Well Concept



@eqm separation  $r_0$  :

$$F_{\text{Attractive}} + F_{\text{Repulsive}} = 0$$

Terms of energies:

$$E_{\text{net}} = E_{\text{Attractive}} + E_{\text{Repulsive}}$$

$$E_{\text{net}} = -\frac{A}{r} + \frac{B}{r^n} \quad n=7-12$$

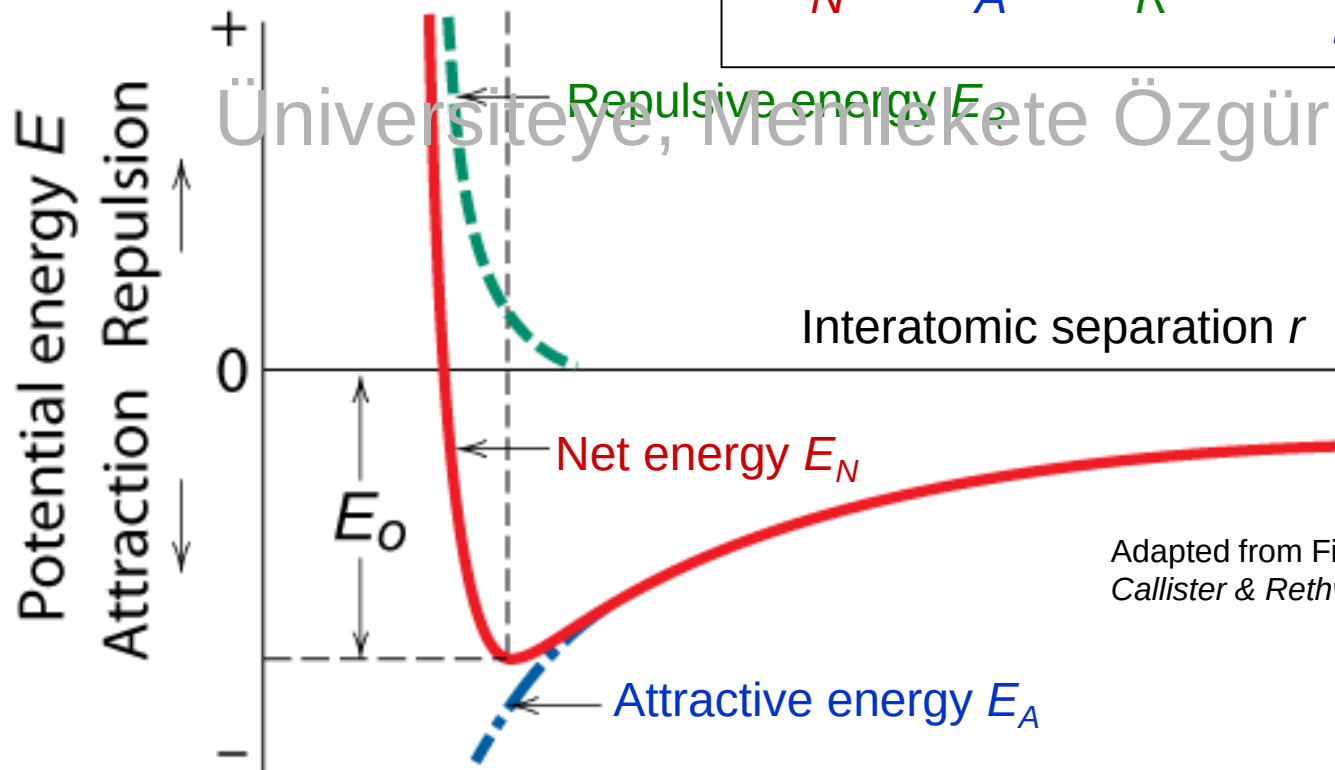
Coulomb attraction  
(*very long range*)

Born repulsion  
(*very short range*)

# Ionic Bonding

- Energy – minimum energy most stable
  - Energy balance of **attractive** and **repulsive** terms

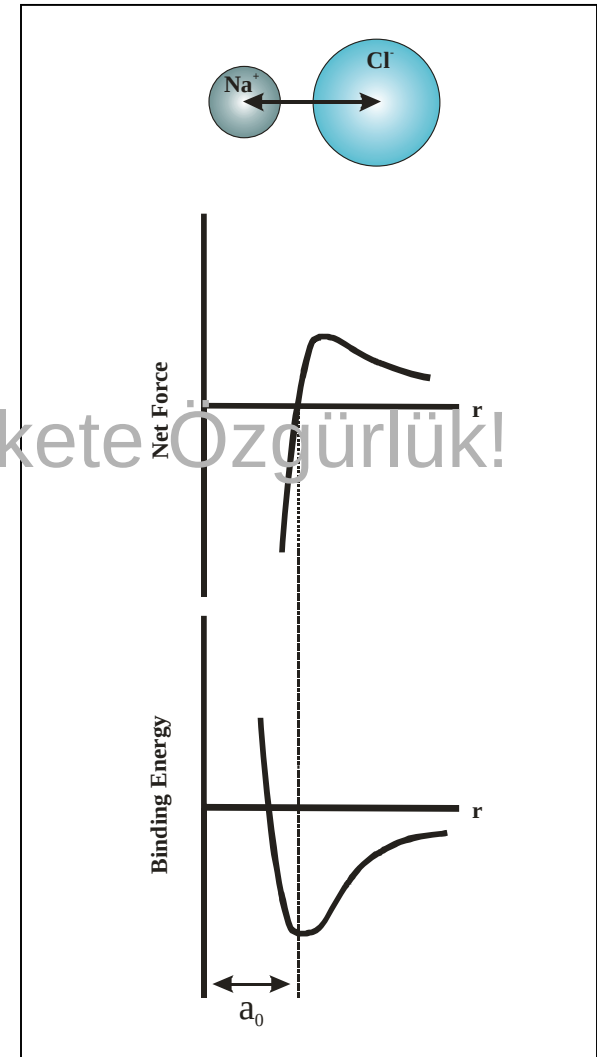
$$E_N = E_A + E_R = -\frac{A}{r} + \frac{B}{r^n}$$



Adapted from Fig. 2.8(b),  
Callister & Rethwisch 8e.

# Ionic Bonding Energy

- **Bonding force is derivative of bonding energy ( $F = -dE/dr$ )**
- **Equilibrium position given by  $F=0$  or  $E=\text{minimum}$**
- **Information available:**
  - **Binding energy**
  - **Interatomic spacing**
  - **Modulus of elasticity**
  - **Coefficient of thermal expansion**



# Examples: Ionic Bonding

- Predominant bonding in **Ceramics**

Diagram illustrating the periodic table with arrows indicating the formation of ionic compounds from metal and non-metal elements:

- NaCl**: Sodium (Na) from Group IA and Chlorine (Cl) from Group VIIA.
- MgO**: Magnesium (Mg) from Group IIA and Oxygen (O) from Group VIA.
- CaF<sub>2</sub>**: Calcium (Ca) from Group IIA and Fluorine (F) from Group VIIA.
- CsCl**: Cesium (Cs) from Group IA and Chlorine (Cl) from Group VIIA.

|           |           |                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |          |          |          |          |         |
|-----------|-----------|------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----------|----------|----------|----------|---------|
| IA        |           |                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 0        |          |          |          |         |
| H<br>2.1  |           |                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           | He<br>-  |          |          |          |         |
| Li<br>1.0 | Be<br>1.5 |                  |           |           |           |           |           |           |           |           |           |           |           |           |           | B<br>2.0  | C<br>2.5 | N<br>3.0 | O<br>3.5 | F<br>4.0 | Ne<br>- |
| Na<br>0.9 | Mg<br>1.2 | III A            | IV A      | V A       | VI A      | VII A     |           |           |           |           |           |           |           |           |           |           | Ar<br>-  |          |          |          |         |
| K<br>0.8  | Ca<br>1.0 | Sc<br>1.3        | Ti<br>1.5 | V<br>1.6  | Cr<br>1.6 | Mn<br>1.5 | Fe<br>1.8 | Co<br>1.8 | Ni<br>1.8 | Cu<br>1.9 | Zn<br>1.6 | Ga<br>1.6 | Ge<br>1.8 | As<br>2.0 | Se<br>2.4 | Br<br>2.8 | Kr<br>-  |          |          |          |         |
| Rb<br>0.8 | Sr<br>1.0 | Y<br>1.2         | Zr<br>1.4 | Nb<br>1.6 | Mo<br>1.8 | Tc<br>1.9 | Ru<br>2.2 | Rh<br>2.2 | Pd<br>2.2 | Ag<br>1.9 | Cd<br>1.7 | In<br>1.7 | Sn<br>1.8 | Sb<br>1.9 | Te<br>2.1 | I<br>2.5  | Xe<br>-  |          |          |          |         |
| Cs<br>0.7 | Ba<br>0.9 | La-Lu<br>1.1-1.2 | Hf<br>1.3 | Ta<br>1.5 | W<br>1.7  | Re<br>1.9 | Os<br>2.2 | Ir<br>2.2 | Pt<br>2.2 | Au<br>2.4 | Hg<br>1.9 | Tl<br>1.8 | Pb<br>1.8 | Bi<br>1.9 | Po<br>2.0 | At<br>2.2 | Rn<br>-  |          |          |          |         |
| Fr<br>0.7 | Ra<br>0.9 | Ac-No<br>1.1-1.7 |           |           |           |           |           |           |           |           |           |           |           |           |           |           |          |          |          |          |         |

← Give up electrons

→ Acquire electrons

Adapted from Fig. 2.7, Callister & Rethwisch 8e. (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.

# Covalent Bonding

- **Cooperative sharing of valence electrons between two adjacent atoms**
- **Very directional**
- **Atoms want to have their valence shell filled**
  - For s and p subshells, this means sharing 8 electrons
  - Hydrogen and the transition metals are exceptions
- **Example: Chlorine**
  - 7 valence electrons
  - $\text{Cl}_2$  forms by electron sharing

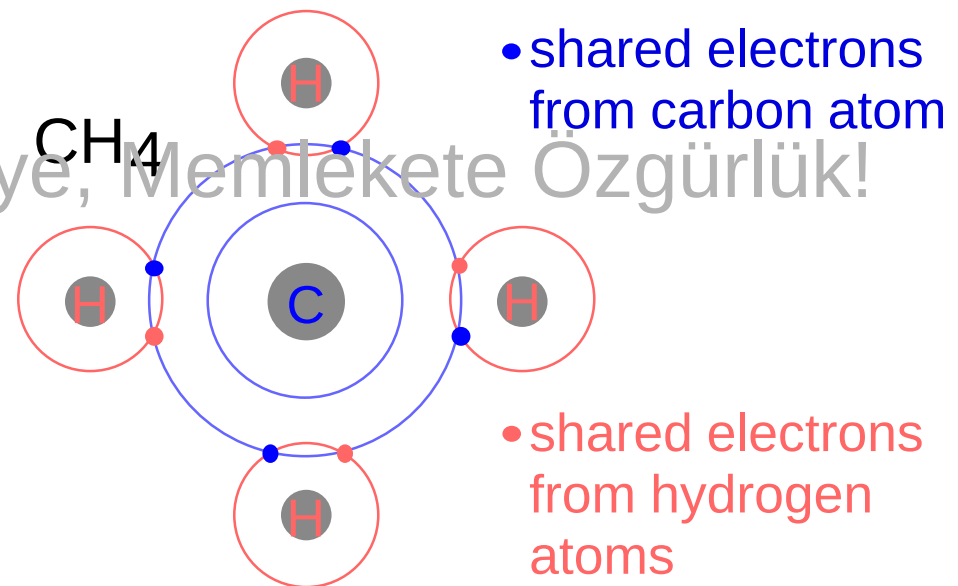
# Covalent Bonding

- similar **electronegativity**  $\therefore$  share electrons
- bonds determined by valence – s & p orbitals dominate bonding
- Example: CH<sub>4</sub>

C: has 4 valence e<sup>-</sup>,  
needs 4 more

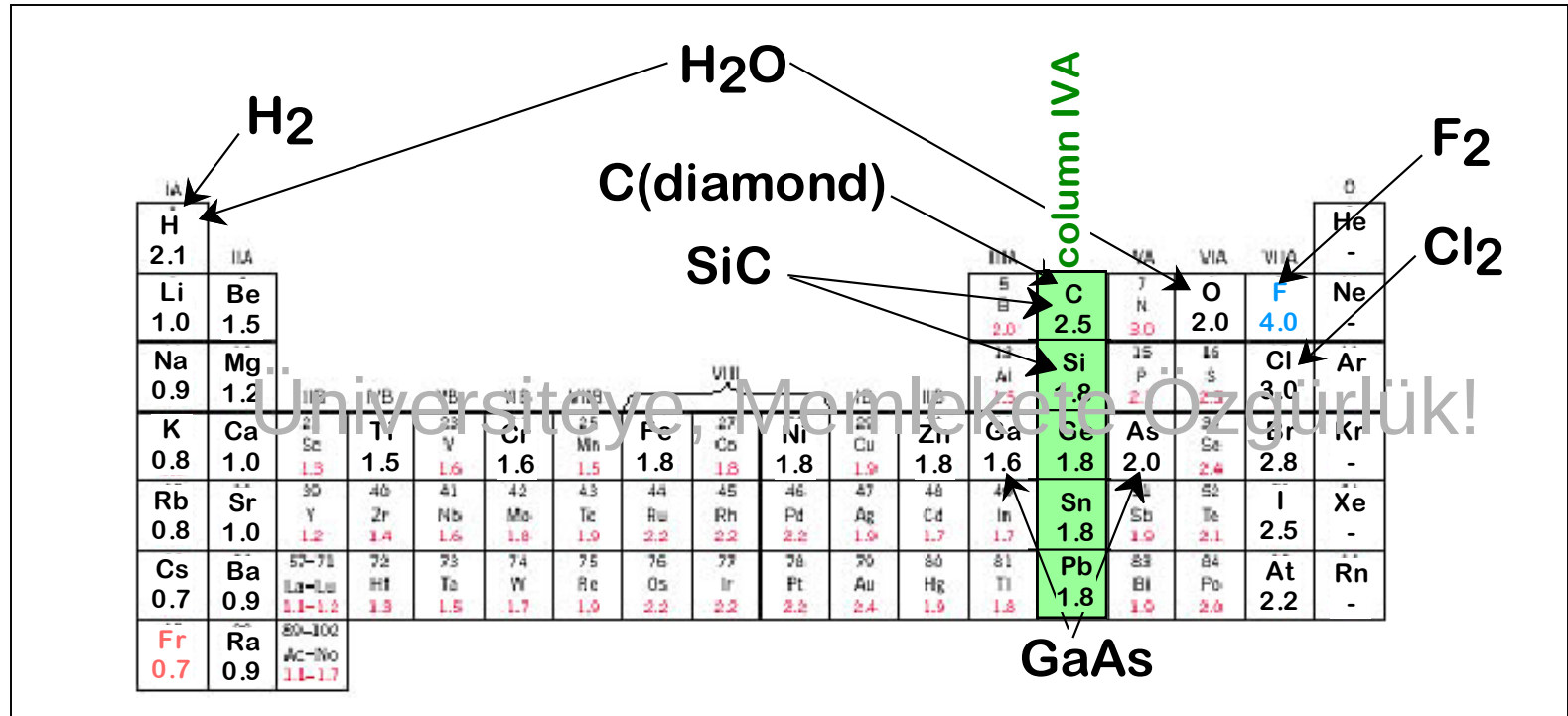
H: has 1 valence e<sup>-</sup>,  
needs 1 more

Electronegativities  
are comparable.



Adapted from Fig. 2.10, Callister & Rethwisch 8e.

# Examples of Covalent Bonding



The diagram shows a periodic table with electronegativity values. Arrows point from specific elements to examples of covalent bonding:

- H<sub>2</sub>**: Points to Hydrogen (H) in Group 1A.
- H<sub>2</sub>O**: Points to Oxygen (O) in Group 6A.
- C(diamond)**: Points to Carbon (C) in Group 4A.
- SiC**: Points to Silicon (Si) in Group 4A.
- F<sub>2</sub>**: Points to Fluorine (F) in Group 7A.
- Cl<sub>2</sub>**: Points to Chlorine (Cl) in Group 7A.
- GaAs**: Points to Gallium (Ga) in Group 3A and Arsenic (As) in Group 5A.

A green vertical bar highlights **column IVA**, which includes Carbon (C), Silicon (Si), and Germanium (Ge).

|           |           |                   |           |           |           |           |           |           |           |           |           |           |           |           |           |           |         |
|-----------|-----------|-------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|---------|
| 1A        | 2A        | VIII              |           |           |           |           |           |           |           |           |           | 3A        | 4A        | 5A        | 6A        | 7A        | 8A      |
| H<br>2.1  | He<br>-   | Li<br>1.0         | Be<br>1.5 | B<br>2.0  | C<br>2.5  | N<br>3.0  | O<br>2.0  | F<br>4.0  | Ne<br>-   | Na<br>0.9 | Mg<br>1.2 | Al<br>1.5 | Si<br>1.8 | P<br>2.2  | S<br>2.5  | Cl<br>3.0 | Ar<br>- |
| K<br>0.8  | Ca<br>1.0 | Sc<br>1.3         | Ti<br>1.5 | V<br>1.6  | Cr<br>1.6 | Mn<br>1.5 | Fe<br>1.8 | Co<br>1.8 | Ni<br>1.8 | Cu<br>1.9 | Zn<br>1.8 | Ga<br>1.6 | Ge<br>1.8 | As<br>2.0 | Se<br>2.4 | Br<br>2.8 | Kr<br>- |
| Rb<br>0.8 | Sr<br>1.0 | Y<br>1.2          | Zr<br>1.4 | Nb<br>1.6 | Mo<br>1.8 | Tc<br>1.9 | Ru<br>2.2 | Rh<br>2.2 | Pd<br>2.2 | Ag<br>1.9 | Cd<br>1.7 | In<br>1.7 | Sn<br>1.8 | Sb<br>1.9 | Te<br>2.1 | I<br>2.5  | Xe<br>- |
| Cs<br>0.7 | Ba<br>0.9 | La-Lu<br>1.1-1.2  | Hf<br>1.3 | Ta<br>1.5 | W<br>1.7  | Re<br>1.9 | Os<br>2.2 | Ir<br>2.2 | Pt<br>2.2 | Au<br>2.4 | Hg<br>1.9 | Tl<br>1.8 | Pb<br>1.8 | Bi<br>1.9 | Po<br>2.0 | At<br>2.2 | Rn<br>- |
| Fr<br>0.7 | Ra<br>0.9 | Ac-100<br>1.1-1.7 |           |           |           |           |           |           |           |           |           |           |           |           |           |           |         |

- Molecules with non-metals
- Molecules between metals & nonmetals
- Elemental solids on RHS of Periodic Table
- Compound solids about column IVA

# Primary Bonding

- **Metallic Bond** -- delocalized as electron cloud
- **Ionic-Covalent Mixed Bonding**

$$\% \text{ ionic character} = \left( 1 - e^{-\frac{(X_A - X_B)^2}{4}} \right) \times (100\%)$$

where  $X_A$  &  $X_B$  are Pauling electronegativities

Ex: MgO

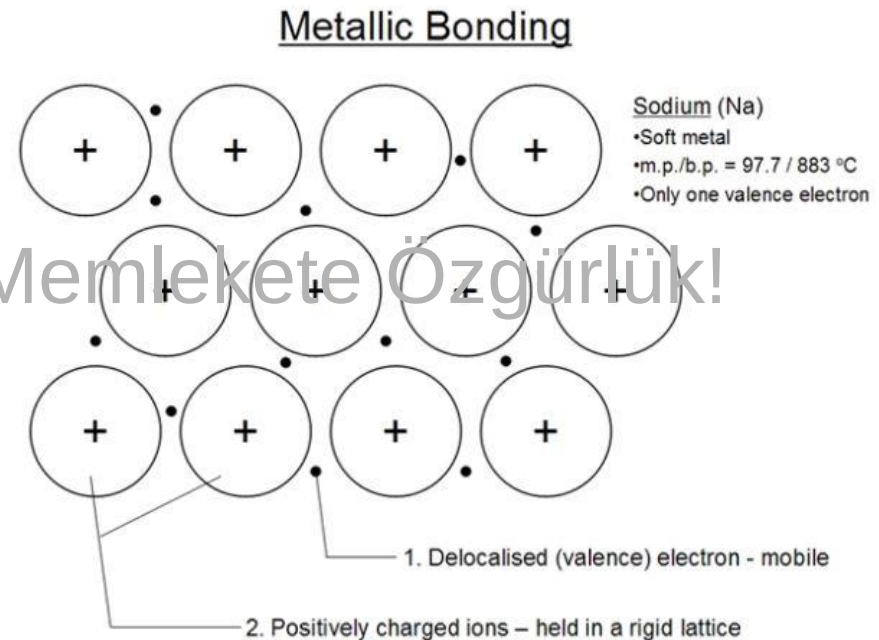
$$\begin{aligned} X_{\text{Mg}} &= 1.2 \\ X_{\text{O}} &= 3.5 \end{aligned}$$

$$\% \text{ ionic character} = \left( 1 - e^{-\frac{(3.5 - 1.2)^2}{4}} \right) \times (100\%) = 73.4\% \text{ ionic}$$



# Metallic Bonding

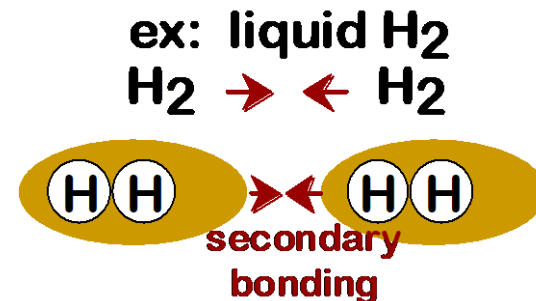
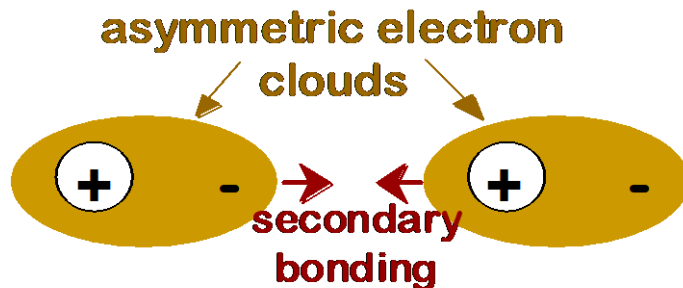
- 1/2/3 valance electrons are not bound to any particular atom in the solid
- Arises from a sea of donated valence electrons to create an electron sea (sometimes called electron cloud and an electron gas)
- Primary bond for metals and alloys (IA, IIA and for all elemental metals)
- Nondirectional bonding (Free e-shield ion cores from mutually repulsive electrostatic forces)
- Bonding may be weak or strong
- High electrical & thermal conductivity



*Atomic (ion) cores (nucleus + core or nonvalence electrons) submerged in an electron gas (outer electrons)*

# Secondary Bonding

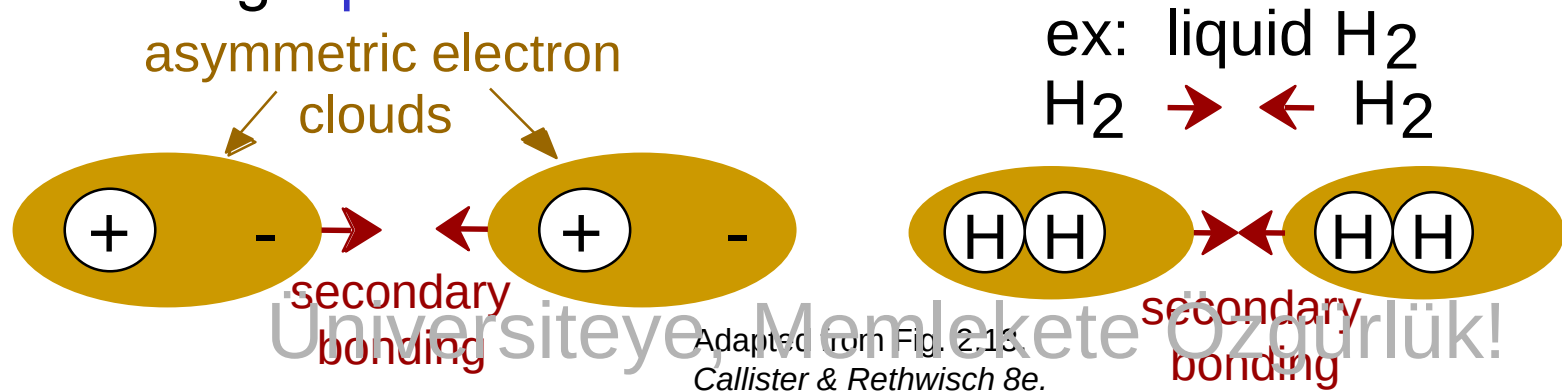
- Bonding without electron transfer or sharing
  - Attraction based on secondary interactions (attractive forces between electric dipoles):
    - Dipole-dipole coupling
    - Dipole-induced dipole interactions
    - Induced dipole-induced dipole interactions
  - Tends to have low melting points
- Fluctuating dipoles



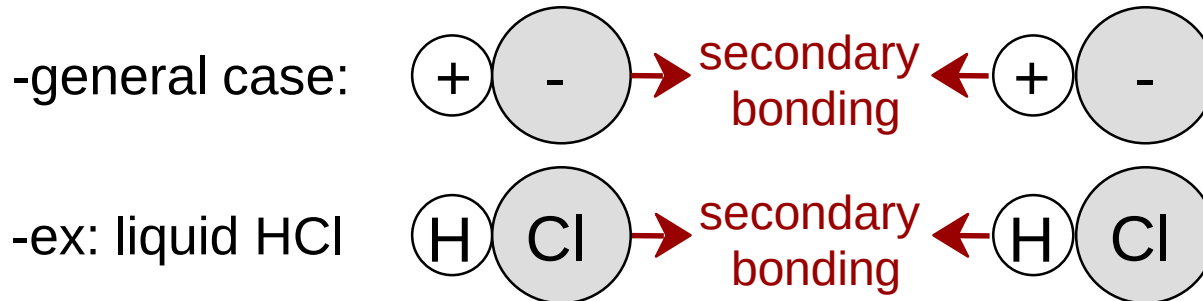
# SECONDARY BONDING

Arises from interaction between **dipoles**

- Fluctuating **dipoles**



- Permanent **dipoles**-molecule induced



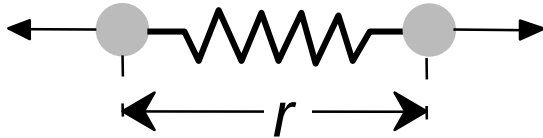
Adapted from Fig. 2.15, Callister & Rethwisch 8e.



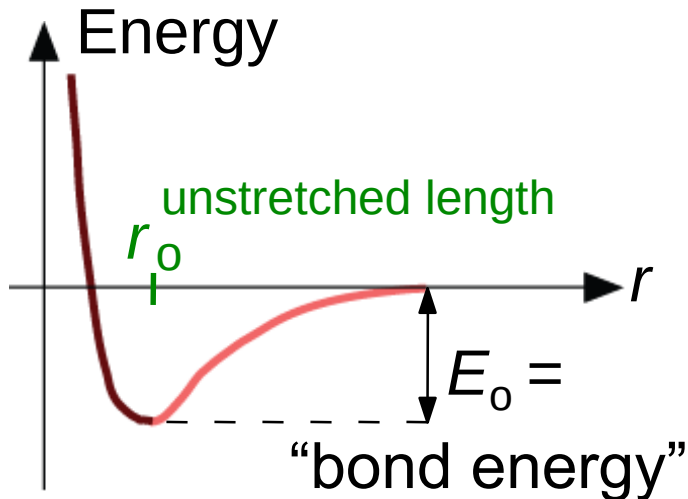
secondary bonding

# Properties From Bonding: $T_m$

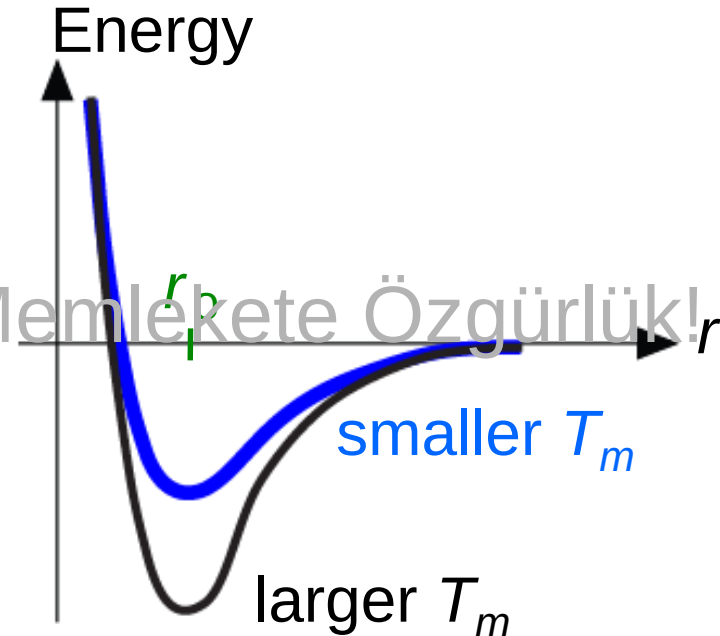
- Bond length,  $r$



- Bond energy,  $E_0$



- Melting Temperature,  $T_m$



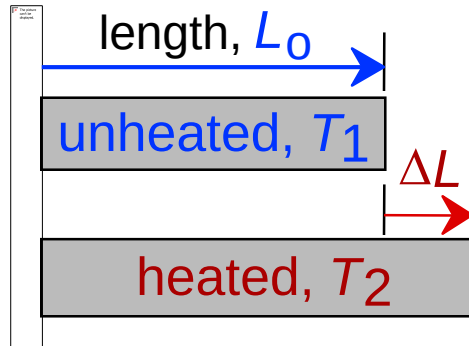
$T_m$  is larger if  $E_0$  is larger.

# Comparison of Melting Points

| Material         | Bonding Type         | Melting Point (°C) |
|------------------|----------------------|--------------------|
| NaCl             | Ionic                | 801                |
| Diamond          | Covalent             | 3550               |
| $(C_2H_4)_n$     | Covalent & Secondary | 120                |
| Cu               | Metallic             | 1085               |
| Ar               | Secondary            | -189               |
| H <sub>2</sub> O | Secondary            | 0                  |

# Properties From Bonding : $\alpha$

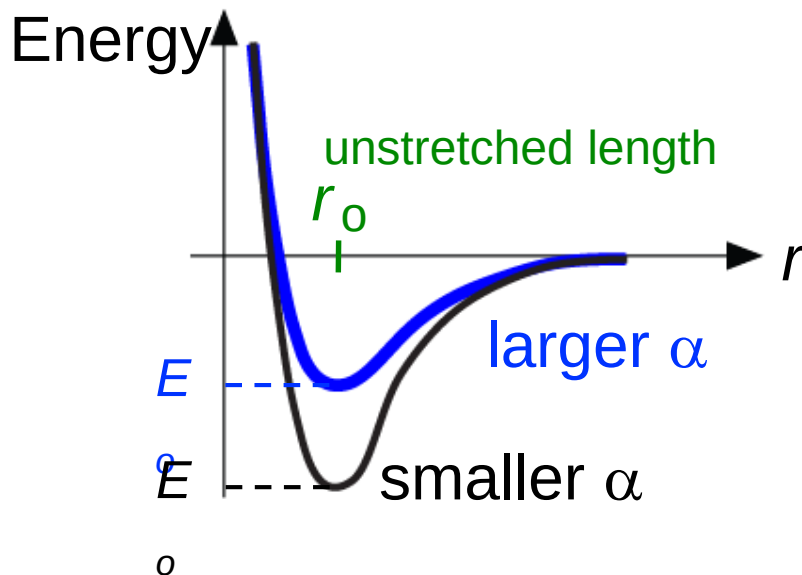
- Coefficient of thermal expansion,  $\alpha$



coeff. thermal expansion

$$\frac{\Delta L}{L_0} = \alpha (T_2 - T_1)$$

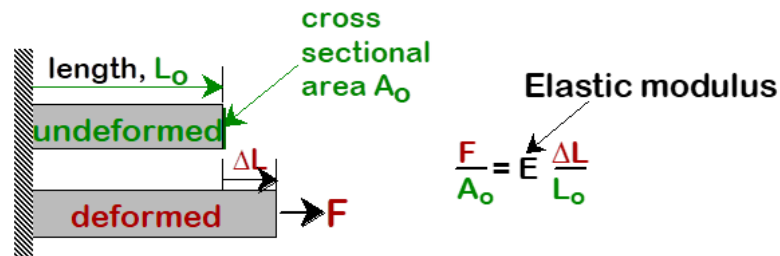
- $\alpha \sim$  symmetric at  $r_0$



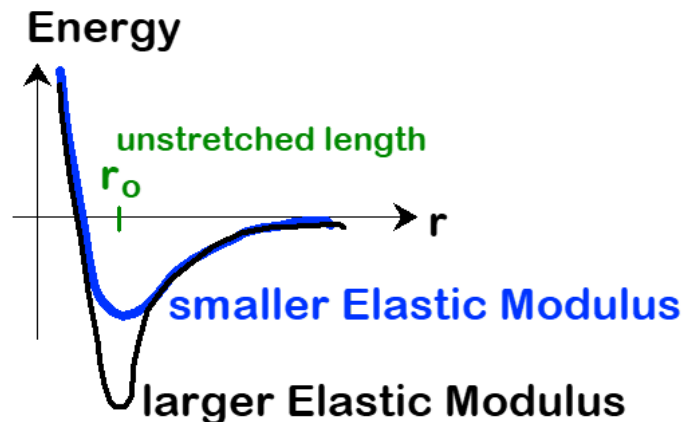
$\alpha$  is larger if  $E_0$  is smaller.

# Properties From Bonding : $E$

- Elastic modulus

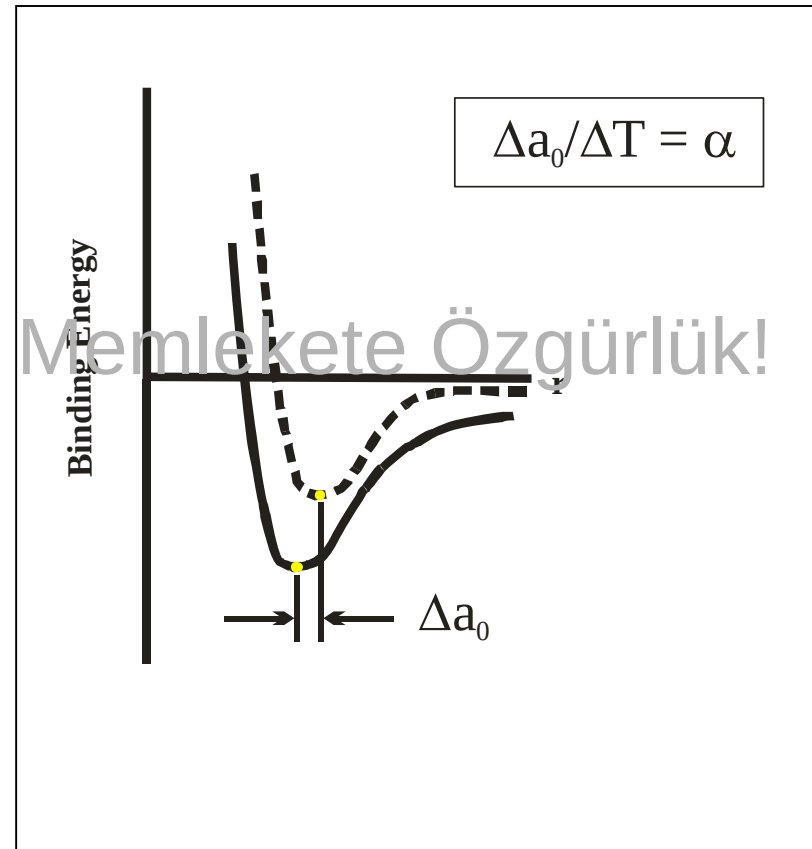


Üniversiteye, Memlekete Özgürlük!



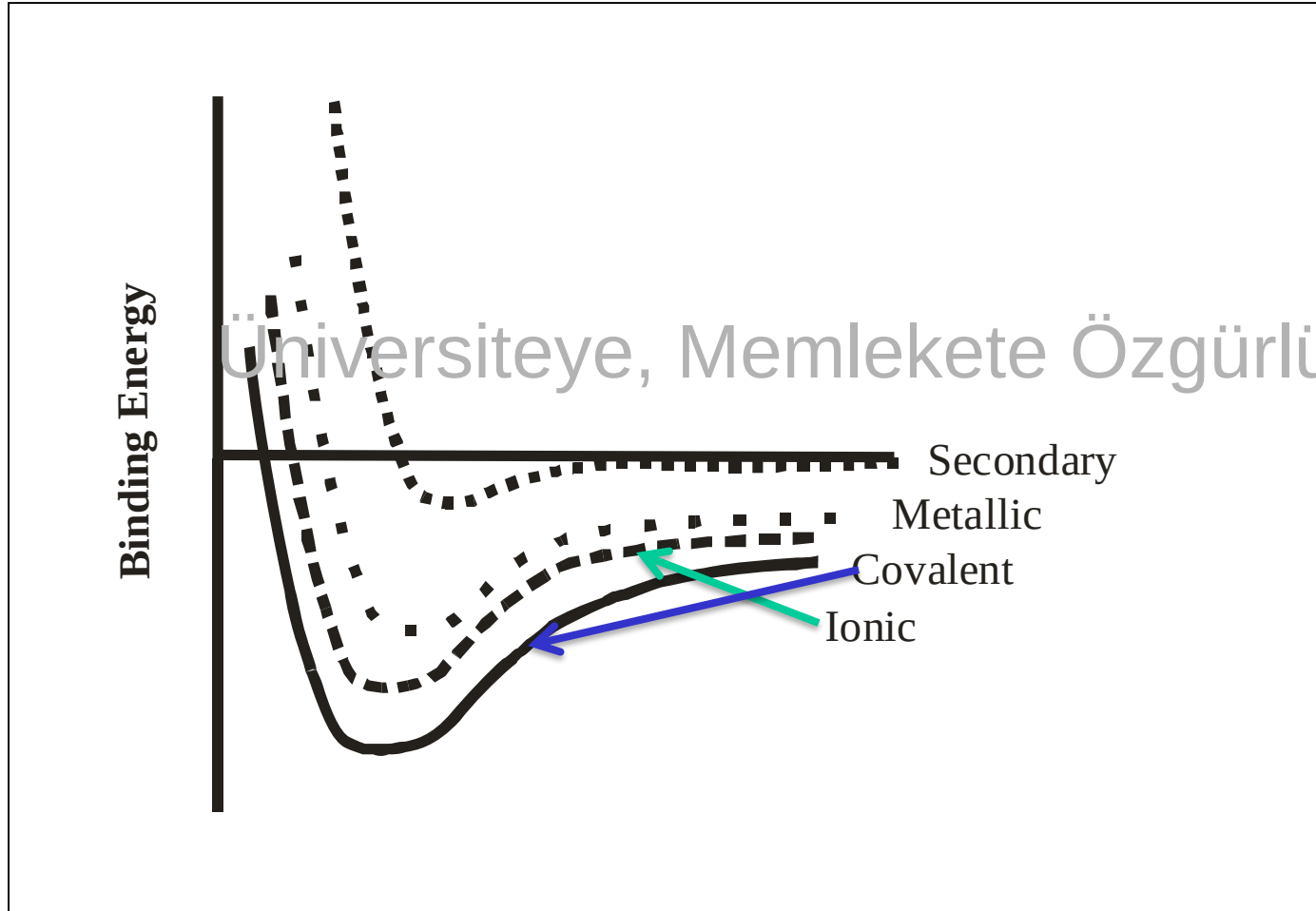
# Temperature Dependence

- Binding energy curve is strongly dependent on temperature
- Shift in equilibrium position gives thermal expansion coefficient
- Slope of curve is related to Young's Modulus

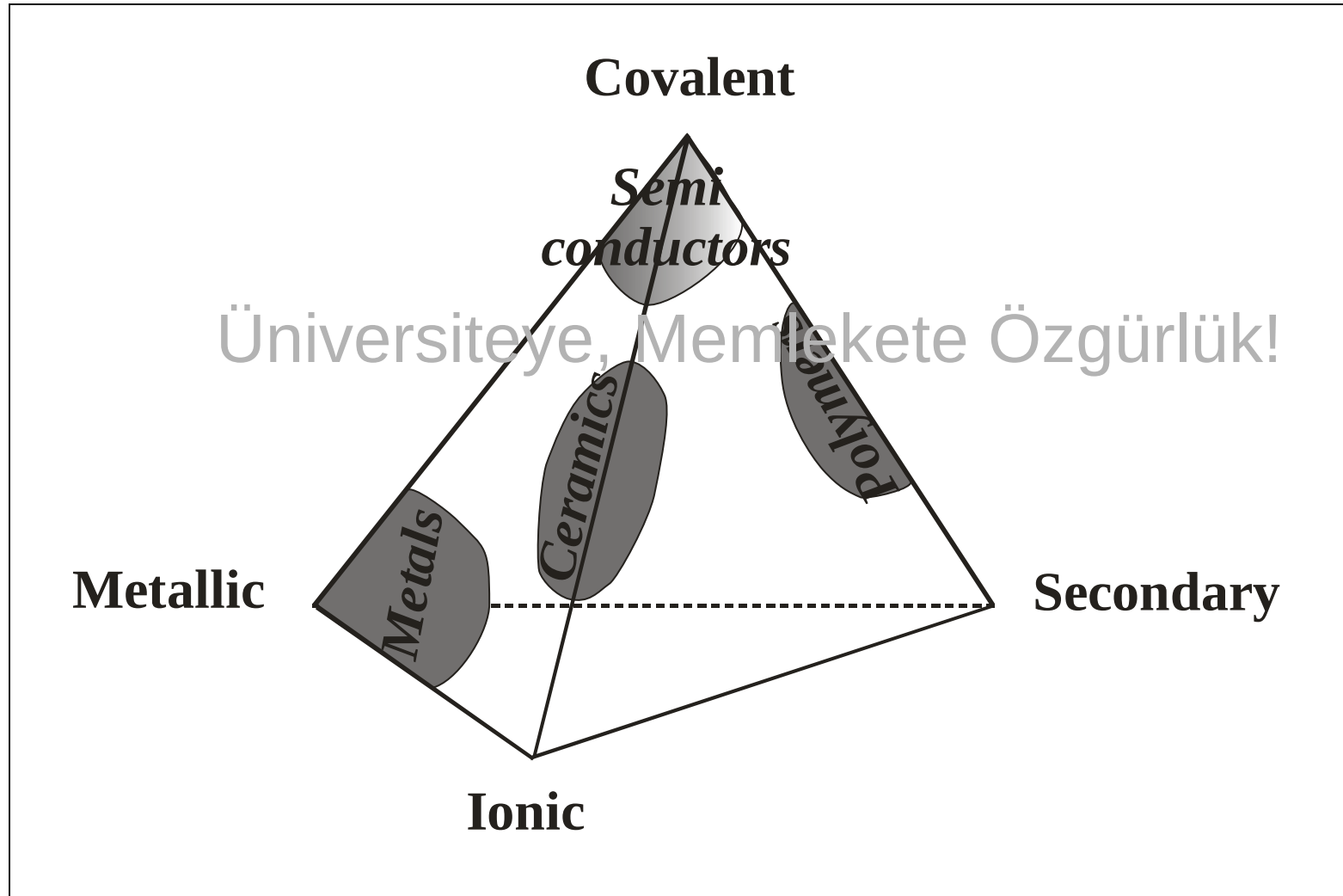




# Relative Strengths of Bonding Types



# Bonding in Materials



# Summary: Primary Bonds

## Ceramics

(Ionic & covalent bonding):

Large bond energy

large  $T_m$

large  $E$

small  $\alpha$

## Metals

(Metallic bonding):

Variable bond energy

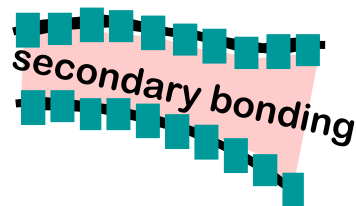
moderate  $T_m$

moderate  $E$

moderate  $\alpha$

## Polymers

(Covalent & Secondary):



Directional Properties

Secondary bonding dominates

small  $T_m$

small  $E$

large  $\alpha$

# Summary: Chemical Bonding

| <u>Type</u> | <u>Bond Energy</u>                          | <u>Comments</u>                                                                    |
|-------------|---------------------------------------------|------------------------------------------------------------------------------------|
| Ionic       | Large!                                      | Nondirectional ( <b>ceramics</b> )                                                 |
| Covalent    | Variable<br>large-Diamond<br>small-Bismuth  | Directional<br>( <b>semiconductors</b> , <b>ceramics</b><br><b>polymer</b> chains) |
| Metallic    | Variable<br>large-Tungsten<br>small-Mercury | Nondirectional ( <b>metals</b> )                                                   |
| Secondary   | smallest                                    | Directional<br>inter-chain ( <b>polymer</b> )<br>inter-molecular                   |